

Table 1: Hit ratio at round 5 across nine BDA benchmark datasets (5 seeds, mean). A = Coreset (algorithmic), B = LLM Reasoning (pure LLM), C = Waddington (online ML + LLM + routing). Bold indicates the best value per row.

Dataset	A (Coreset)	B (LLM)	C (Waddington)
IFN- γ	0.100	0.162	0.193
IL-2	0.139	0.265	0.343
Sanchez21	0.031	0.064	0.091
Sanchez21 (down)	0.078	0.068	0.096
Carnevale22	0.054	0.057	0.057
Scharenberg22	0.286	0.461	0.441
Steinhart	0.090	0.154	0.149
K562-Essential	0.270	0.527	0.575
K562-GWPS	0.156	0.219	0.339
Average	0.134	0.220	0.254

Waddington (C) outperforms the pure-LLM baseline (B) on 6 of 9 datasets and matches it on 2 (Scharenberg22, Steinhart), achieving a +15% relative gain in average hit ratio (0.254 vs. 0.220). C surpasses the algorithmic baseline (A) on all 9 datasets (+90% relative). The two near-ties (Scharenberg22, Steinhart) correspond to narrow, well-characterized biological tasks where LLM parametric priors are already near-optimal and additional ML adaptation offers limited marginal benefit.

Table 2: Ablation study: hit ratio at round 5 (5 seeds, mean). C is the full Waddington arm. C – Mem removes cross-experiment memory from the LLM prompt. C – LLM removes the LLM entirely (online ML only). C – ML freezes ML at the leave-one-out prior (no in-experiment retraining). Shuffled replaces all gene names passed to the LLM with anonymous identifiers (GENE_00001, ...). Bold marks the highest value per row; underline marks the lowest.

Dataset	C	C – Mem	C – LLM	C – ML	Shuffled
IFN- γ	0.193	0.189	0.190	<u>0.179</u>	0.175
IL-2	0.343	0.346	0.349	0.355	<u>0.351</u>
Sanchez21	0.091	0.091	0.098	<u>0.077</u>	0.080
Sanchez21 (down)	0.096	0.097	0.093	<u>0.093</u>	0.092
Carnevale22	0.057	0.059	<u>0.051</u>	0.064	0.059
Scharenberg22	0.441	0.478	0.592	<u>0.433</u>	0.551
Steinhart	0.149	0.156	0.090	0.149	<u>0.076</u>
K562-Essential	0.575	0.559	0.476	<u>0.527</u>	0.524
K562-GWPS	0.339	0.343	0.343	<u>0.309</u>	0.343
Average	0.254	0.258	0.254	<u>0.243</u>	0.250

Removing cross-experiment memory (C – Mem) has no measurable effect ($\Delta = +0.002$), indicating that the LLM’s parametric knowledge subsumes the information provided via the memory prompt in this leave-one-out benchmark setup. Removing the LLM (C – LLM) costs -0.005 on average but incurs large dataset-specific losses: Steinhart (-0.063 , -47% relative) and K562-Essential (-0.108 , -19% relative), where narrow biological pathways are well-encoded in LLM priors but not

in PPI or DepMap features. Conversely, Scharenberg22 improves without LLM (+0.123) because the LLM’s CAR-T gene knowledge conflicts with the strong ML signal (leave-one-out AUC 0.825 on v3 DepMap features). Freezing ML at the LOO prior (C – ML) produces the largest consistent degradation (−0.015 average, −6%), confirming that in-experiment retraining is the primary driver of C’s advantage over static methods. Anonymizing gene names (Shuffled) selectively collapses performance on pathway-specific tasks (Steinhart −0.069, K562-Essential −0.044) while improving Scharenberg22 (+0.078), demonstrating that LLM contributions on those tasks are entirely mediated by biological gene-name semantics rather than by task context or experimental feedback.